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TELUGU DOCUMENT WRITER ASSITANT

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A Grammarly-like tool for Telugu

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Project Motivation & Problem

Problem Statement

- ▶ Limited Telugu NLP Tools vs English
- ▶ Digital divide in rural/educational contexts
- ▶ Agglutinative morphology (complex word forms)
- ▶ Need for grammatically correct, context-aware Telugu documents
- ▶ Preserve and promote Telugu in digital age

Target

- ▶ Improve Telugu digital literacy
- ▶ Reduce writing time & errors
- ▶ Support students & institutions (Telugu medium)
- ▶ Enable professional / government / academic content creation

Literature Review

Core NLP Concepts

Language Modeling :

N-Gram Models → Predictive text completion

Spell Correction Techniques:

Levenshtein Distance → Spell correction using edit distance

Hybrid Approaches:

Rule-based + Morphological analysis

Key Research Papers

- Rule Based Pseudo N-Gram Model for Telugu Script
- Predicting Sentences using N-Gram Language Models
- Spell Correction and Suggestion Using Levenshtein Distance
- Spelling Correction of OCR-generated Hindi Text Using BERT and Levenshtein Distance

Rule Based Pseudo N-Gram Model for Telugu Script

Pros:

- Handles Telugu morphology
- Extracts root words via suffix removal
- Useful for spell check & completion

Cons:

- Hard to generalize (rule-based)
- Focused on root words, not full assistant

Predicting Sentences using N-Gram Language Models

Pros:

- High precision
- Works well on predictable/low-entropy text
- Can reduce keystrokes by up to 60%

Cons:

- Limited context
- Poor performance on diverse/unstructured text

Spell Correction and Suggestion Using Levenshtein Distance

Pros:

- Simple, effective, easy to implement
- Optimal edit distance (≤ 2) works well for typos

Cons:

- Context-free $\rightarrow$ ignores sentence meaning
- Tested on small domain, scalability not shown

Spelling Correction of OCR-generated Hindi Text Using BERT and Levenshtein Distance

Pros:

- Works well for morphologically rich languages (Hindi, Telugu)
- Detects spelling errors with minimal edits
- Effective for generating close candidate words

Cons:

- Ignores context $\rightarrow$ may miss correct suggestions
- Cannot handle real-word errors (valid word, wrong usage)

Core Features

Spell Checker

- Real-time error detection and correction
- Context-aware suggestions
- Support for common Telugu typing errors
- Integration with Telugu keyboard layouts

Sentence Completion

- Predictive text generation based on context
- Domain-specific completions
- Grammar-aware suggestions
- Coherent multi-word completions

Assistant Interface

- User-friendly web application
- Real-time assistance while typing
- Export functionality (PDF, Word formats)
- Performance metrics and accuracy reporting

Data Collection and Corpus

Primary Datasets:

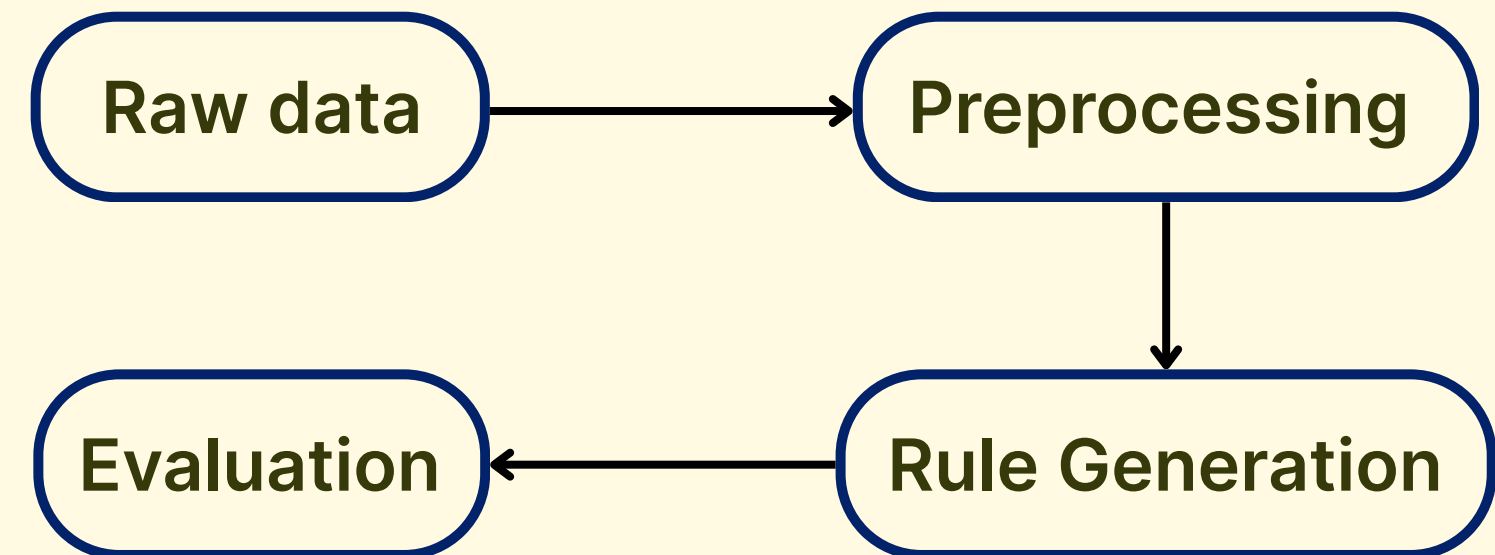
- Telugu Wikipedia Dump (200MB+ text)
- Telugu News Articles
- Literary Collections and Government Documents

Synthetic Data Generation:

- Common misspelling patterns (keyboard layouts)
- Phonetic error simulation
- Incomplete sentence datasets (for completion models)

Data Preprocessing Pipeline:

- Text normalization and cleaning.
- Tokenization for the Telugu script.
- Error pattern extraction for training / evaluation



Methodology and Workflow

Sentence Completion

Baseline:

N-grams + Hidden Markov Models

- Simple, interpretable
- Limitation: Data sparsity, computationally heavy

Enhanced:

LSTM / Transformer models

- Capture long-range dependencies
- Better context understanding

Spell Correction

Basic:

Levenshtein Edit Distance

- Dictionary-based, fast, reliable for typos

Advanced:

Neural / classification models

- Context-aware corrections
- Handles complex errors

Integration & Evaluation

Unified pipeline:

- Combine spell correction + sentence completion

Evaluation Metrics:

- Perplexity, Accuracy, User Satisfaction

Future Scope

- Grammar & style checking
- Predictive text
- Multi-language support

Application Development

- Build interactive Telugu Writing Assistant (GUI / Web App)
- Demonstrate practical utility and real-world applicability

Thankyou